Assignment #3 - Solutions

```
library(spdep)
library(tidyr)
library(ggplot2)
library(gridExtra)
library(MASS)
library(sf)
library(dplyr)
library(readr)
library(stringr)
library(tigris)
```

1)

Refer to Figure 1 (*Poverty by Socio-Economic Factors in Wisconsin/Illinois in 2000*) from *Poverty in Wisconsin and Illinois Data Example Exploratory Data Analysis*. Interpret what you observe in the visualization for the other seven socio-economic factors.

Percent working in agriculture is between 8% and 16.31% by the summary statistics. As agriculture increases, poverty does not change much. Percent retail is centered around about 10% with not much relation to poverty. Poverty appears to decrease as manufacturing increases, but then levels off. With percent foreign, there appears to be a slow and steady leveling off of poverty as foreign increases. As high school education increases, poverty appears to remain stable until a drop at about 37% hsch. For percent Black, we see that as poverty increases, so does percent Black. However, for percent Hispanic, poverty remains fairly flat after a decrease at about 2% Hispanic.

2)

Refer to *Poverty in Wisconsin and Illinois Data Example Simple Linear Regression*.

(a) Complete the interpretation for all other regression coefficients for `m1`

Our results from the summary output above show us that the fitted intercept is 34.11866. The fitted regression coefficient for `ag` is 0.12953. The p-value is 0.200814. There is no evidence of a relationship between `ag` and `povty` after accounting for all other socio-economic factors. The fitted regression coefficient for `manu` is -0.15246 and has a p-value of 0.000644. There is very strong evidence of a relationship between `manu` and `povty` after accounting for all other socio-economic factors. For `retail`, the fitted regression coefficient is -0.09620, with a standard error of 0.13363 and p-value of 0.472612. There is no evidence of a relationship between `retail` and `povty` after accounting for all other socio-economic factors. The fitted regression coefficient for `foreign` is 0.36065 and the p-value is 0.098084. There is weak evidence of a relationship between `foreign` and `povty` after accounting for all other socio-economic factors. The fitted regression coefficient for `hsch` is -0.10548 and the p-value is 0.137235. There is no evidence of a relationship between `hsch` and `povty` after accounting for all other socio-economic factors. The fitted regression coefficient for `black` is 0.18226, the standard error is 0.04865, and the p-value is 0.000248. There is very strong evidence of a relationship between `black` and `povty` after accounting for all other socio-economic factors. Lastly, the fitted regression coefficient for `hisp` is -0.46182 and the p-value is 0.008652. There is strong evidence of a relationship between `hisp` and `povty` after accounting for all other socio-economic factors.

(b) Complete the interpretation for all other regression coefficients for `m1_back`

From the model summary output above, the fitted regression coefficient for `manu` is -0.17211 and the p-value is less than 2×10^{-16}. This indicates that a 1% increase in county-level manufacturing rate is associated with a 0.172% decrease in poverty, after accounting for all other socio-economic factors. The fitted coefficient for `foreign` is 0.43641 and the p-value is 0.03344. A 1% increase in county-level foreign-born rate is associated with a 0.436% increase in the poverty rate, after accounting for all other socio-economic factors. The fitted regression for `black` is 0.19796 and the p-value is 2.33×10^{-5}. A 1% increase in county-level proportion Black is associated with a 0.198% increase in poverty, after accounting for all other socio-economic factors. Lastly, the fitted regression coefficient for `hisp` is -0.49005 and the p-value is 0.00518. A 1% increase in county-level proportion Hispanic is associated with a 0.49% decrease in poverty, after accounting for all other socio-economic factors.

(c) Perform the stepwise regression using `direction="both"`. How do the results change, if any?

```
library(sf)
library(spdep)

ilwipov_shape <- st_read("ilwipov_shp.shp", quiet = TRUE)

ilwipov_nb <- poly2nb(ilwipov_shape, row.names = ilwipov_shape$FIPS_n)
listw_povW <- nb2listw(ilwipov_nb, style = "W")

m1_min <- lm(povty ~ 1, data = ilwipov_shape)

m1_both <- step(
  m1_min,
  direction = "both",
  scope = (~ ag + manu + retail + foreign + feemp + hsch + black + hisp)
)
```

```
## Start:  AIC=500.27
## povty ~ 1
##
##           Df Sum of Sq    RSS    AIC
## + feemp    1    862.76 2186.6 444.40
## + manu     1    733.69 2315.7 454.38
## + black    1    483.13 2566.2 472.26
## + hsch     1    165.33 2884.0 492.57
## <none>                 3049.3 500.27
## + hisp     1     26.08 3023.3 500.78
## + retail   1      7.35 3042.0 501.85
## + ag       1      6.51 3042.8 501.90
## + foreign  1      6.15 3043.2 501.92
##
## Step:  AIC=444.4
## povty ~ feemp
##
##           Df Sum of Sq    RSS    AIC
## + manu     1    374.27 1812.3 413.73
## + black    1    254.85 1931.7 424.84
## + hsch     1    219.15 1967.4 428.02
## <none>                 2186.6 444.40
```

```
## + foreign  1     14.88 2171.7 445.21
## + ag       1     11.69 2174.9 445.47
## + hisp     1      2.94 2183.7 446.17
## + retail   1      1.06 2185.5 446.32
## - feemp    1    862.76 3049.3 500.27
##
## Step:  AIC=413.73
## povty ~ feemp + manu
##
##           Df Sum of Sq    RSS    AIC
## + black    1    153.40 1658.9 400.35
## + hsch     1     53.95 1758.4 410.48
## <none>                  1812.3 413.73
## + retail   1     14.44 1797.9 414.34
## + foreign  1      4.84 1807.5 415.27
## + ag       1      2.40 1809.9 415.50
## + hisp     1      1.26 1811.0 415.61
## - manu     1    374.27 2186.6 444.40
## - feemp    1    503.34 2315.7 454.38
##
## Step:  AIC=400.35
## povty ~ feemp + manu + black
##
##           Df Sum of Sq    RSS    AIC
## + hisp     1     38.17 1620.7 398.30
## <none>                  1658.9 400.35
## + ag       1      8.56 1650.3 401.45
## + foreign  1      6.00 1652.9 401.72
## + retail   1      5.85 1653.1 401.73
## + hsch     1      3.25 1655.7 402.00
## - black    1    153.40 1812.3 413.73
## - manu     1    272.82 1931.7 424.84
## - feemp    1    408.60 2067.5 436.66
##
## Step:  AIC=398.3
## povty ~ feemp + manu + black + hisp
##
##           Df Sum of Sq    RSS    AIC
## + foreign  1     43.18 1577.6 395.60
## + hsch     1     23.65 1597.1 397.74
## <none>                  1620.7 398.30
## + retail   1      7.91 1612.8 399.44
## + ag       1      0.86 1619.9 400.20
## - hisp     1     38.17 1658.9 400.35
## - black    1    190.31 1811.0 415.61
## - manu     1    247.46 1868.2 421.02
## - feemp    1    356.15 1976.9 430.86
##
## Step:  AIC=395.6
## povty ~ feemp + manu + black + hisp + foreign
##
##           Df Sum of Sq    RSS    AIC
## <none>                  1577.6 395.60
## + hsch     1      8.02 1569.5 396.71
```

```
## + retail   1      5.56 1572.0 396.98
## + ag       1      4.91 1572.7 397.05
## - foreign  1     43.18 1620.7 398.30
## - hisp     1     75.35 1652.9 401.72
## - black    1    177.90 1755.5 412.19
## - manu     1    188.44 1766.0 413.23
## - feemp    1    399.26 1976.8 432.85
```

```
summary(m1_both)
```

```
##
## Call:
## lm(formula = povty ~ feemp + manu + black + hisp + foreign, data = ilwipov_shape)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3269 -1.8913 -0.1027  1.2358 13.3191
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 29.59209    2.60749  11.349  < 2e-16 ***
## feemp       -0.31457    0.04824  -6.521 7.87e-10 ***
## manu        -0.17211    0.03842  -4.480 1.38e-05 ***
## black        0.19796    0.04548   4.353 2.33e-05 ***
## hisp        -0.49005    0.17300  -2.833  0.00518 **
## foreign      0.43641    0.20351   2.144  0.03344 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.064 on 168 degrees of freedom
## Multiple R-squared:  0.4827, Adjusted R-squared:  0.4673
## F-statistic: 31.35 on 5 and 168 DF,  p-value: < 2.2e-16
```

The final model selected is the same as using backward elimination.

3)

Consider the following grid of experimental plots below. The data in `corn.csv` consist of organic phosphorus in corn (y), organic phosphorus in soil ($x1$), and inorganic phosphorus in soil ($x2$) for each of the plots in the grid. Consider the following models:

- I: $y = \beta_0 + \beta_1 x_1 + \varepsilon$
- II: $y = \beta_0 + \beta_2 x_2 + \varepsilon$
- III: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30

For each model:

(a) Perform exploratory spatial data analysis on the data

```
library(spdep)
library(ggplot2)

corn <- read.csv("corn.csv")
set.seed(1)

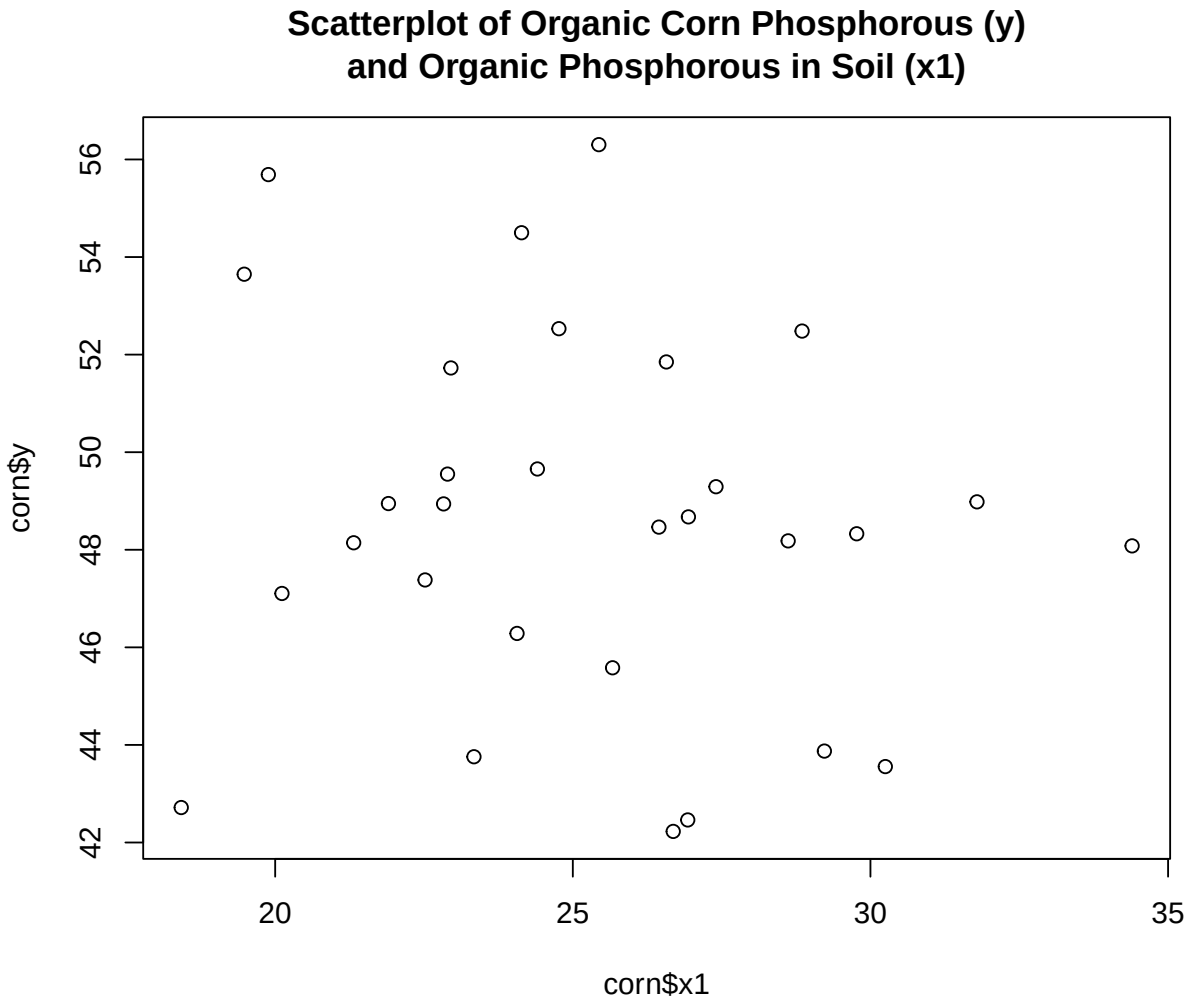
# create neighbourhood structure object
corn.nb <- cell2nb(nrow = 6, ncol = 5, type = "rook")
matrix(seq(1:30), ncol = 5, byrow = TRUE)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    2    3    4    5
## [2,]    6    7    8    9   10
## [3,]   11   12   13   14   15
## [4,]   16   17   18   19   20
## [5,]   21   22   23   24   25
## [6,]   26   27   28   29   30
```

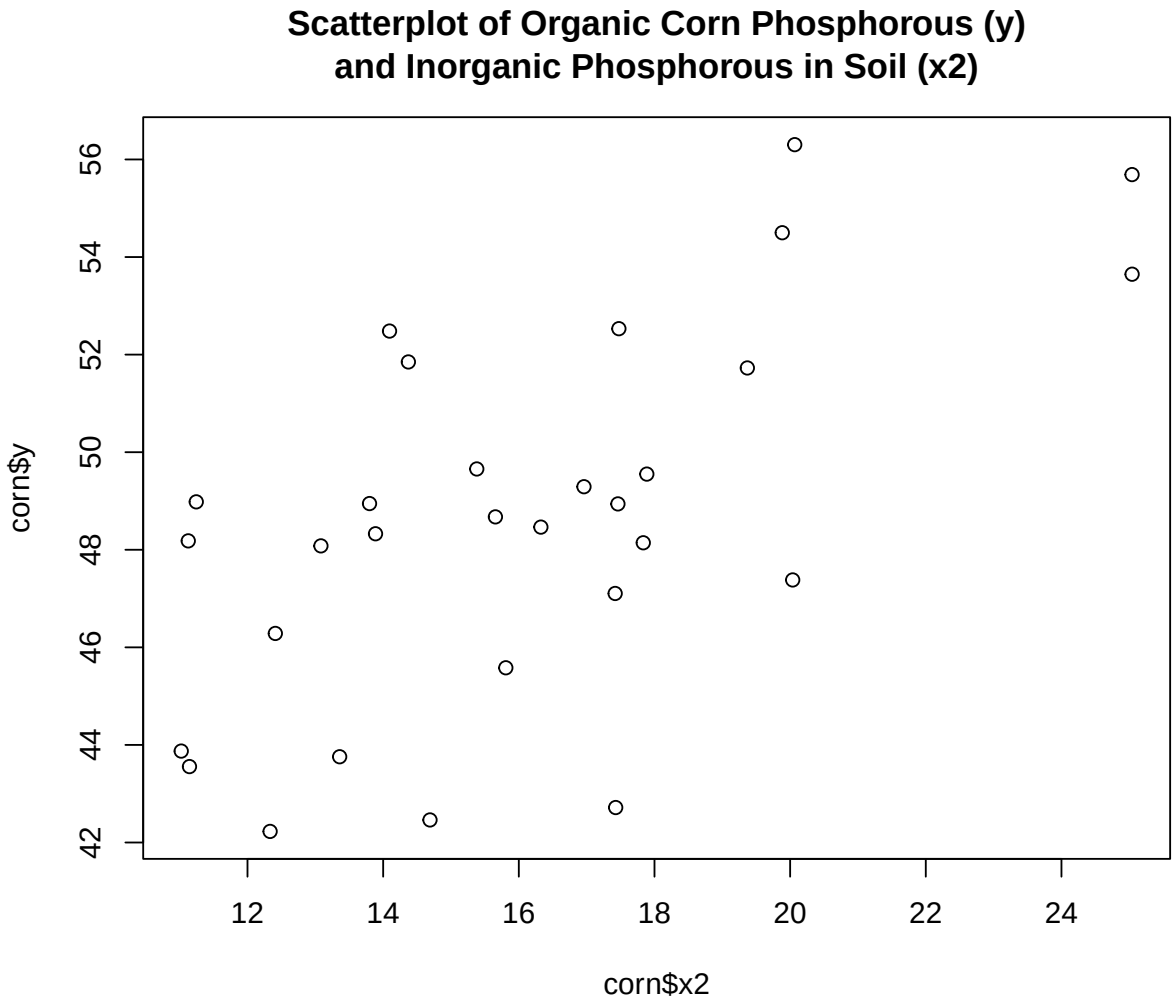
```
corn.lw <- nb2listw(corn.nb, style = "B")
```

ESDA using base R

```
plot(x = corn$x1, y = corn$y,
     main = "Scatterplot of Organic Corn Phosphorous (y)\nand Organic Phosphorous in Soil (x1)")
```




```
plot(x = corn$x2, y = corn$y,
     main = "Scatterplot of Organic Corn Phosphorous (y)\nand Inorganic Phosphorous in Soil (x2)")
```

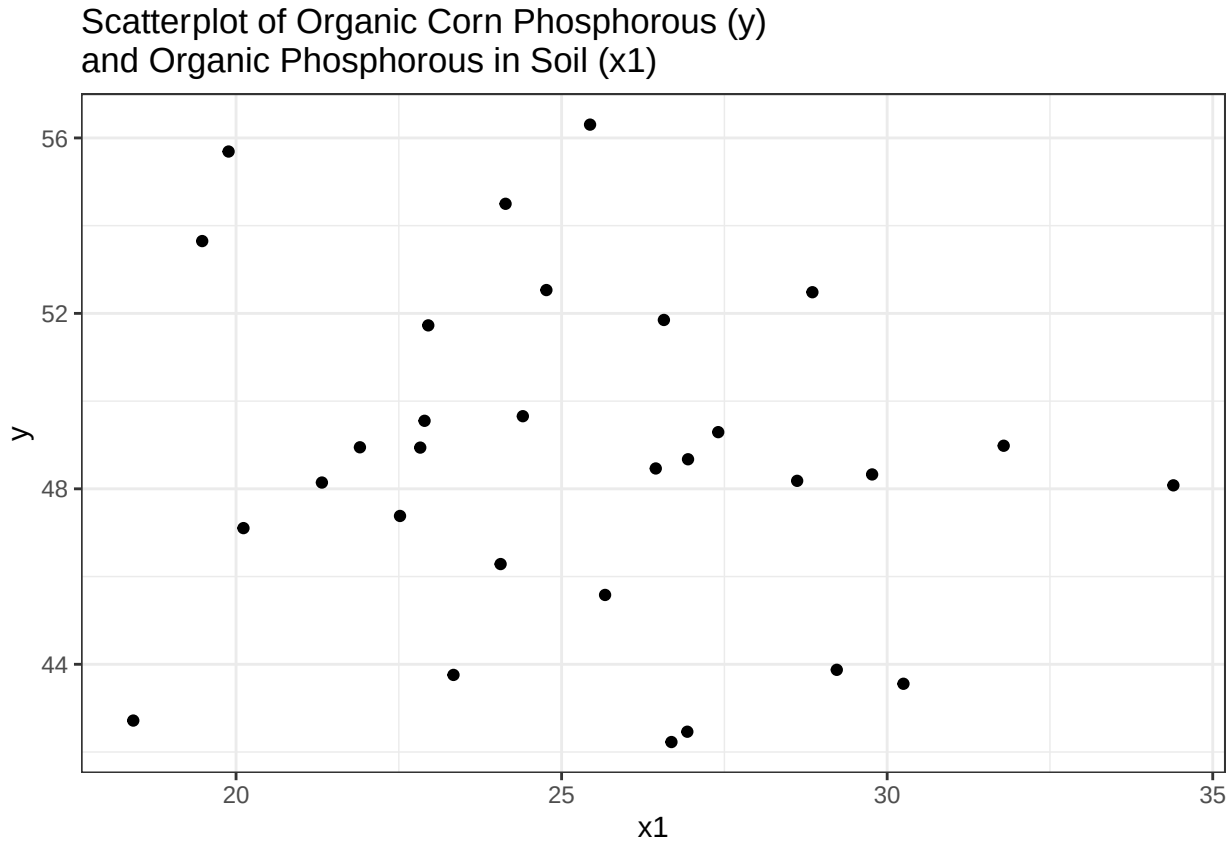


Using ggplot2

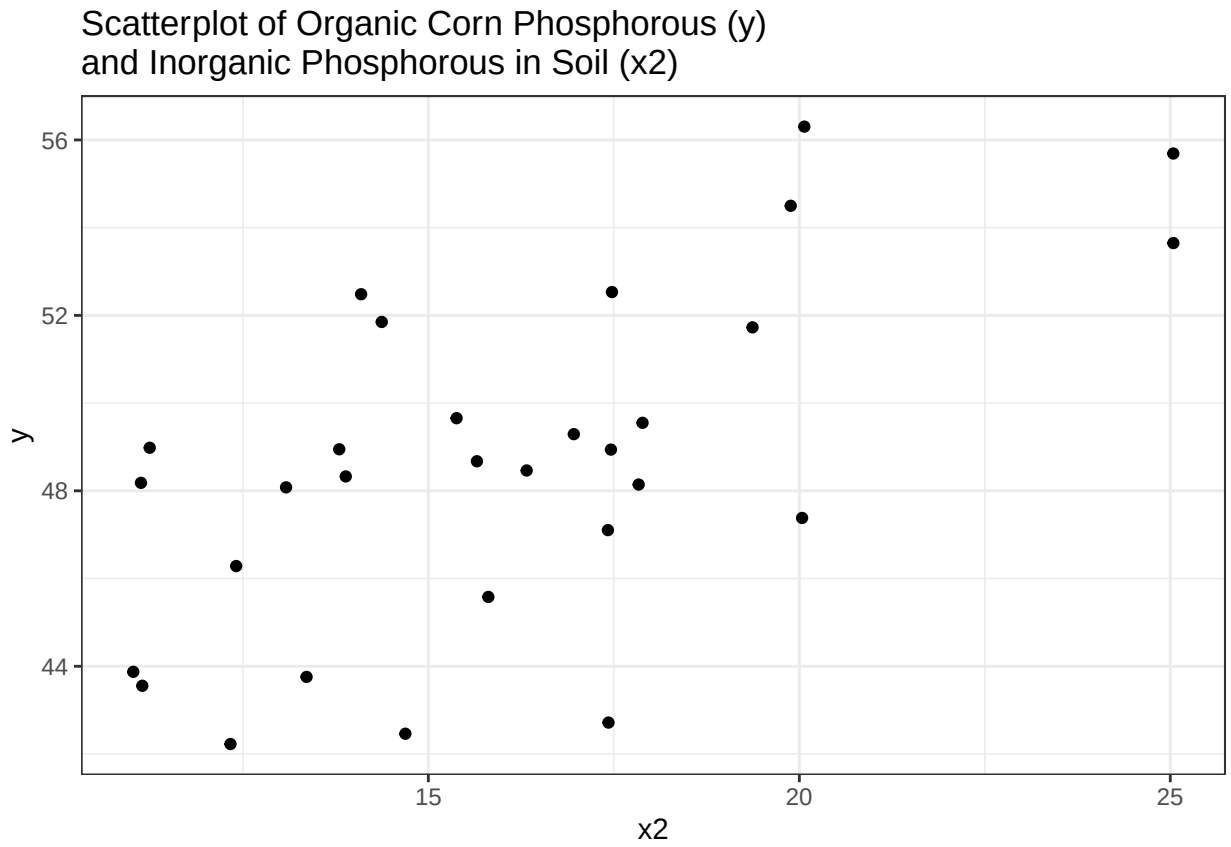
```
p1 <- ggplot(corn, aes(x = x1, y = y)) +
  geom_point() +
  theme_bw() +
  ggtitle("Scatterplot of Organic Corn Phosphorous (y)\nand Organic Phosphorous in Soil (x1)")

p2 <- ggplot(corn, aes(x = x2, y = y)) +
  geom_point() +
  theme_bw() +
  ggtitle("Scatterplot of Organic Corn Phosphorous (y)\nand Inorganic Phosphorous in Soil (x2)")

p1
```


Scatterplot of Organic Corn Phosphorous (y)
and Organic Phosphorous in Soil (x1)
56
52
48
44
y
20
25
30
35
x1

```
p2
```



(Note: You may as well instead include other visualizations that address the spatial relationships in the data.)

It is difficult to say anything about x_1 and y. There appears to be a linear relationship between x_2 and y.

(b) Fit the models using the usual regression approach (ignoring spatial considerations). Construct and interpret a residual plot

```
# model I
m1 <- lm(y ~ x1, data = corn)

# model II
m2 <- lm(y ~ x2, data = corn)

# model III
m3 <- lm(y ~ x1 + x2, data = corn)
```

Model I residual plots

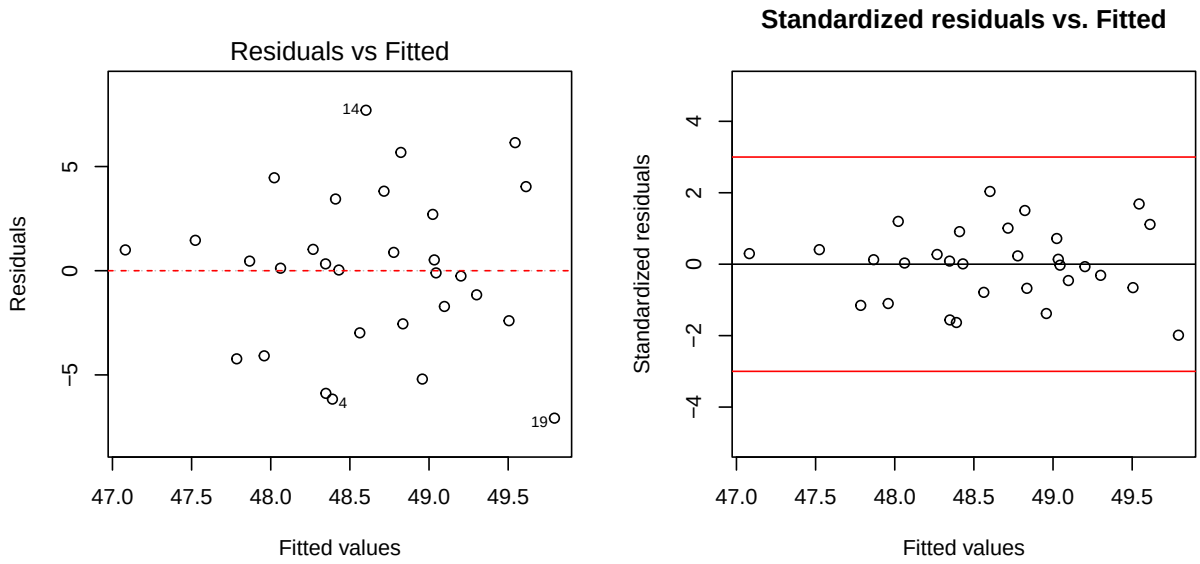
```
par(mfrow = c(2, 2))

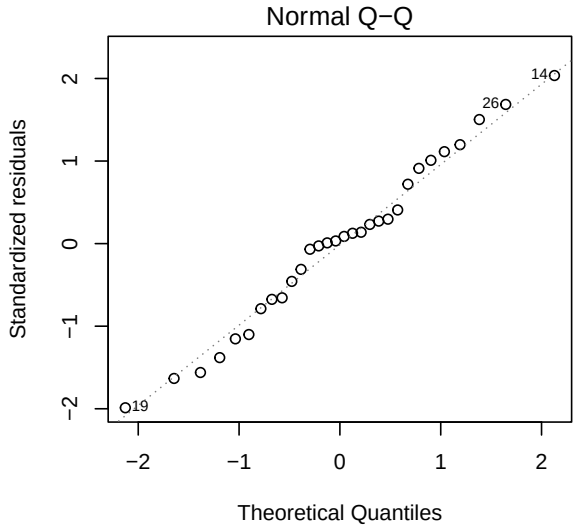
# Residuals vs. fitted
plot(m1, which = 1, add.smooth = FALSE)
abline(h = 0, col = "red", lty = 2)

stdresids <- stdres(m1)
plot(x = fitted(m1), y = stdresids,
```

```
     xlab = "Fitted values",
     ylab = "Standardized residuals",
     main = "Standardized residuals vs. Fitted",
     ylim = c(-5, 5))
abline(h = c(-3, 0, 3), col = c("red", "black", "red"))

# QQ plot
plot(m1, which = 2, add.smooth = FALSE)
```





No evidence of outliers from standardized residuals vs. fitted plot. However some evidence of residuals as a function of fitted values. QQ plot looks good; normality assumption is reasonable.

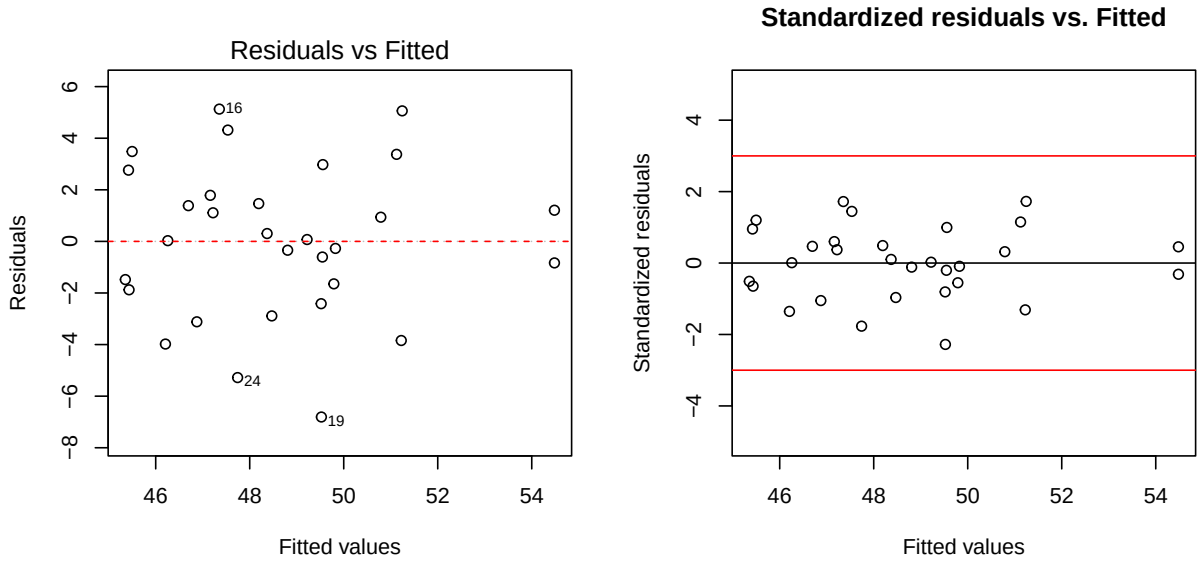
Model II residual plots

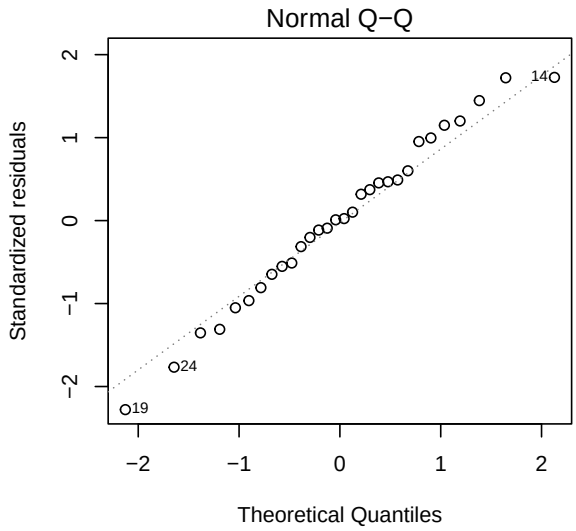
```
par(mfrow = c(2, 2))

plot(m2, which = 1, add.smooth = FALSE)
abline(h = 0, col = "red", lty = 2)

stdresids <- stdres(m2)
plot(x = fitted(m2), y = stdresids,
     xlab = "Fitted values",
     ylab = "Standardized residuals",
     main = "Standardized residuals vs. Fitted",
     ylim = c(-5, 5))
abline(h = c(-3, 0, 3), col = c("red", "black", "red"))

plot(m2, which = 2, add.smooth = FALSE)
```





Residuals look relatively evenly spread around 0 line. Some potential outliers are indicated in the first plot. Based on standardized residuals, we should consider observation #30 as a potential outlier/leverage point and explore it in more detail. In QQ plot, normality assumption is reasonable, with some indication of heavy tails.

Model III residual plots

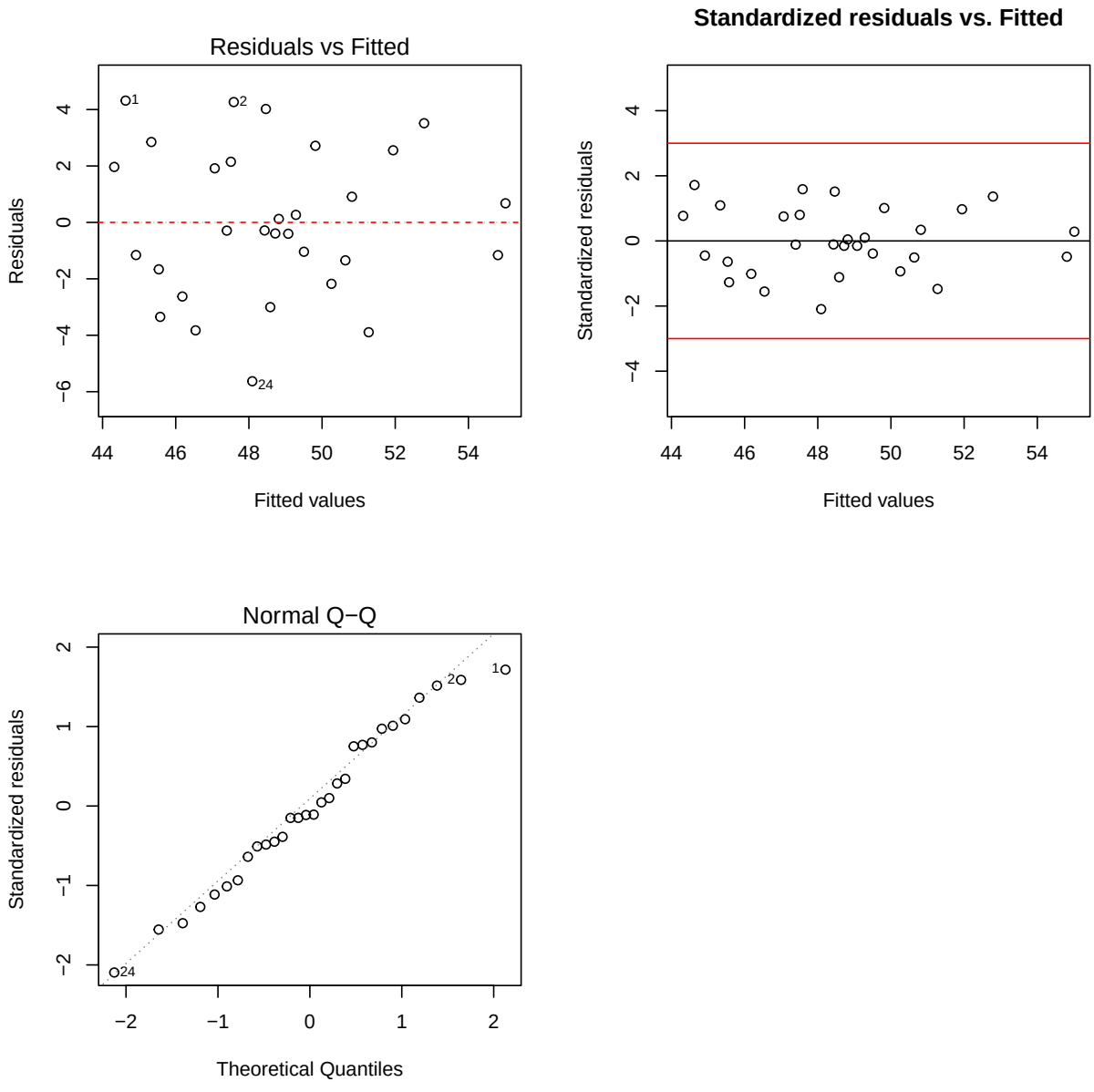
```
par(mfrow = c(2, 2))

plot(m3, which = 1, add.smooth = FALSE)
abline(h = 0, col = "red", lty = 2)

stdresids <- stdres(m3)
plot(x = fitted(m3), y = stdresids,
```

```
     xlab = "Fitted values",
     ylab = "Standardized residuals",
     main = "Standardized residuals vs. Fitted",
     ylim = c(-5, 5))
abline(h = c(-3, 0, 3), col = c("red", "black", "red"))

plot(m3, which = 2, add.smooth = FALSE)
```



Based on first two plots, we should explore observation #30 in more detail. Residuals look okay. QQ plot has more indication of heavy tails.

(c) Perform and interpret test of significance for the slope(s) and intercept for the regression

Interpreting the Wald test on each $\hat{\beta}$ coefficient in each model.

```
summary(m1)
```

```
##
## Call:
## lm(formula = y ~ x1, data = corn)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.0757 -2.5127  0.2242  2.3921  7.7017
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  52.9188     4.7911  11.045 1.03e-11 ***
## x1           -0.1697     0.1876  -0.905    0.373
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.848 on 28 degrees of freedom
## Multiple R-squared:  0.02841,    Adjusted R-squared:  -0.006291
## F-statistic: 0.8187 on 1 and 28 DF,  p-value: 0.3733
```

```
summary(m2)
```

```
##
## Call:
## lm(formula = y ~ x2, data = corn)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -6.8082 -1.8194  0.0497  1.7053  5.1288
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  38.1755     2.5279   15.10 5.51e-15 ***
## x2            0.6512     0.1536    4.24  0.00022 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.046 on 28 degrees of freedom
## Multiple R-squared:  0.391,  Adjusted R-squared:  0.3692
## F-statistic: 17.98 on 1 and 28 DF,  p-value: 0.0002203
```

```
summary(m3)
```

```
##
## Call:
## lm(formula = y ~ x1 + x2, data = corn)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.6297 -1.5840 -0.2886  2.1028  4.3168
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  19.4880     7.1686   2.719   0.0113 *
## x1            0.5086     0.1850   2.749   0.0105 *
## x2            1.0148     0.1913   5.304 1.35e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.742 on 27 degrees of freedom
## Multiple R-squared:  0.5242, Adjusted R-squared:  0.4889
## F-statistic: 14.87 on 2 and 27 DF,  p-value: 4.423e-05
```

In model I, there is very strong evidence that the intercept is not equal to zero; there is no evidence that β_1 is not equal to zero.

In model II, there is weak evidence that the intercept is different from zero. There is very strong evidence that β_2 is different from zero.

In model III, there is weak evidence that the intercept (β_0) is different from zero. There is no evidence that β_1 is not zero, whereas there is very strong evidence that β_2 is different from zero.

4)

For the three models in #3, test spatial correlation among the residuals, assuming a rook's case neighborhood and binary spatial weights. Using `moran.test()`.
Do this part for all 3 models.

Hint: the following R commands may be helpful:

```
corn.lm <- lm(y ~ x1, data = corn)
summary(corn.lm)
plot(corn.lm$fitted, corn.lm$resid)
moran.test(corn.lm$resid, corn.lw, alternative = "two.sided")
```

The `lm` function fits a linear model. In the case above, a linear model of y versus x_1. The `summary` and `anova` functions produce useful output from the regression. Note how we access the residuals and predicted values from the regression using `corn.lm$resid` and `corn.lm$fitted`.

For each model (I, II, III), we will test H_0: CSR vs. H_a: not CSR using Moran's I test.

```
# Model I
moran.test(m1$residuals, corn.lw, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  m1$residuals
## weights: corn.lw
##
## Moran I statistic standard deviate = -0.31424, p-value = 0.7533
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       -0.07690907       -0.03448276        0.01822830
```

```
# Model II
moran.test(m2$residuals, corn.lw, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  m2$residuals
## weights: corn.lw
##
## Moran I statistic standard deviate = 0.34503, p-value = 0.7301
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.01195517       -0.03448276        0.01811434
```

```
# Model III
moran.test(m3$residuals, corn.lw, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  m3$residuals
## weights: corn.lw
##
## Moran I statistic standard deviate = 1.9302, p-value = 0.05358
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.22700098       -0.03448276        0.01835121
```

In summary, for all three models, there is no evidence against CSR in the residuals.

5)

A supposedly better test, although the math is complex, is implemented in `spdep` as follows:

```
lm.morantest(corn.lm, mylw, alternative = "two.sided")
```

Perform this test for the three models in 3, and compare it with the results for 4. Do this part for all 3 models.

```
# Model I
lm.morantest(m1, corn.lw, alternative = "two.sided")
```

```
##
##  Global Moran I for regression residuals
##
## data:
## model: lm(formula = y ~ x1, data = corn)
## weights: corn.lw
##
## Moran I statistic standard deviate = -0.31383, p-value = 0.7536
## alternative hypothesis: two.sided
## sample estimates:
## Observed Moran I      Expectation         Variance
##      -0.07690907      -0.03562449       0.01730541
```

```
# Model II
lm.morantest(m2, corn.lw, alternative = "two.sided")
```

```
##
##  Global Moran I for regression residuals
##
## data:
## model: lm(formula = y ~ x2, data = corn)
## weights: corn.lw
##
## Moran I statistic standard deviate = 0.41285, p-value = 0.6797
## alternative hypothesis: two.sided
## sample estimates:
## Observed Moran I      Expectation         Variance
##       0.01195517      -0.04289288       0.01764930
```

```
# Model III
lm.morantest(m3, corn.lw, alternative = "two.sided")
```

```
##
##  Global Moran I for regression residuals
##
## data:
## model: lm(formula = y ~ x1 + x2, data = corn)
## weights: corn.lw
##
## Moran I statistic standard deviate = 1.9914, p-value = 0.04644
## alternative hypothesis: two.sided
## sample estimates:
## Observed Moran I      Expectation         Variance
##       0.22700098      -0.03682005       0.01755086
```

P-values are slightly different from (c). However, we have the same conclusions that there is no evidence against CSR in the residuals.

6)

Interpret your findings for the problem as a whole using the likelihood ratio test. For the three models in #3, test either x_1 or x_2 by comparing a reduced model to the full model. Hint: the following R commands may be helpful:

```
anova(corn.lm_reduced, corn.lm_full, test = "Chisq")
```

Between Models II and III, we can use the likelihood ratio test to test if we can drop x_1.

```
anova(m2, m3, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: y ~ x2
## Model 2: y ~ x1 + x2
##   Res.Df    RSS Df Sum of Sq Pr(>Chi)
## 1     28 259.86
## 2     27 203.03  1    56.828 0.005977 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There is no evidence that the full model (III) is better than the reduced model (II). Additionally, there is no evidence of residual spatial correlation in either model. We could therefore proceed with the reduced model for any subsequent analysis.

If you compared $m1$ to $m3$:

```
anova(m1, m3, test = "Chisq")
```

```
## Analysis of Variance Table
##
## Model 1: y ~ x1
## Model 2: y ~ x1 + x2
##   Res.Df    RSS Df Sum of Sq  Pr(>Chi)
## 1     28 414.56
## 2     27 203.03  1    211.53 1.134e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Using the likelihood ratio test to test if we can drop x_2, we find that there is strong evidence in favor of x_2 and in keeping the full model.